

Anshuman Singh

Computer Science and Engineering | B.Tech.

National Institute of Technology Patna

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Education

National Institute of Technology Patna

Bachelor of Technology in Computer Science and Engineering | CGPA: 8.30

2024 – 2028

Projects

Medico: Generic Medicine Alternative Finder

April 2025 – June 2025

JavaScript, React, Node.js, Express.js, MongoDB, Redis, Python

– github.com/Rogue-strider/Medico-Project

– medico-project.onrender.com

- Developed a full-stack medicine discovery platform containing **more than 120000 medicine records**, enabling branded-medicine search and discovery of generic alternatives.
- Built an asynchronous Python data-ingestion pipeline using **asyncio and thread pools**, integrated with Node.js and Express.js backend services.
- Integrated **MongoDB** for persistent storage and **Redis** for application-level caching, improving query efficiency and reducing repeated database operations.
- Collaborated with Satyam Jha, Anand Kr. Thakur, and Abhay Singh to develop and deliver the project for **ByteVerse Hackathon 2025**.

CPPDock: Lightweight C++ Container Runtime

January 2026 – Present

C++, Linux Kernel APIs, Namespaces, Cgroups, Docker Architecture

– github.com/anshumansingh0010/cppdock

- Developing a lightweight container runtime in **C++** using Linux kernel primitives to provide process and resource isolation without relying on Docker as the container runtime.
- Implemented PID, network, and filesystem isolation using **Linux namespaces**, along with root filesystem initialization, process management, and isolated execution environments.
- Implemented **Linux cgroups** for resource management and designed a modular container lifecycle supporting creation, execution, resource constraints, and container management.

Stock Sense: AI-Powered Financial Intelligence Platform

April 2026 – Present

React, TypeScript, Fastify, Node.js, MongoDB, Mongoose, OpenAI, Gemini

– github.com/anshumansingh0010/stock_analyzer

– stock-analyzer-deployment.vercel.app

- Developing a full-stack financial intelligence platform for **Nifty 50 stock analysis**, market trends, financial news, portfolio management, and watchlists.
- Integrated **OpenAI and Gemini** for financial question answering and sentiment analysis, combining market data with technical indicators including RSI, MACD, and moving averages.
- Built REST APIs using **Fastify, Node.js, MongoDB, and Mongoose**, implementing JWT authentication, portfolio management, watchlists, configurable price and news alerts, caching, and rate limiting.
- Designed asynchronous news ingestion and LLM request processing using background scheduling and caching to improve application resilience and reduce unnecessary API requests.

Technical Skills

Programming Languages: C++ Python Java JavaScript TypeScript

Frameworks and Libraries: React Node.js Fastify Express.js

Databases: MongoDB Redis

Systems and Infrastructure: Linux Docker Linux Namespaces Cgroups

Developer Tools: Git GitHub VS Code Postman

Positions of Responsibility

Member, Robotics Club

March 2025 – Present

National Institute of Technology Patna